

Bohm-flux objective used for ICP surrogate optimization

1. Local Bohm flux

$$\Gamma_B = n_i c_s$$
$$c_s = \sqrt{eT_e/m_{\text{Ar}^+}}$$

Ion density and electron temperature
from the frozen surrogate

2. Wafer-near distribution

$$\bar{\Gamma}_{B,k} = \frac{\sum_{w_k} r \Gamma_B}{\sum_{w_k} r}$$
$$D_{\max} = \max_k |\bar{\Gamma}_{B,k} / \langle \bar{\Gamma}_B \rangle - 1|$$

10 axial layers
32 equal-area radial bins

3. Constrained optimization

$$\min_{\theta} D_{\max}(\theta)$$
$$\langle n \rangle_W \geq 10^{17} \text{ m}^{-3}$$

$\theta = \{N, PP, PP0, \text{coil geometry}\}$